

```
!pip install -q transformers sentencepiece torch
```

```
from transformers import pipeline

print("Import successful")
```

```
Import successful
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Due to an internal KeyError in the transformers pipeline for translation tasks,
# we are manually loading the model and tokenizer to create a translation function.

model_name = "Helsinki-NLP/opus-mt-en-fr"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

# Define a function that mimics the behavior of a translation pipeline
def translate_en_to_fr(text):
    inputs = tokenizer(text, return_tensors="pt", padding=True, truncation=True)
    # The generate method returns sequences of token ids
    outputs = model.generate(**inputs)
    # Decode the generated token ids back to text
    return tokenizer.decode(outputs[0], skip_special_tokens=True)

# Assign the translation function to the variable name the user intended
translator_fr = translate_en_to_fr

print(f"Translator for {model_name} initialized successfully.")
```

```
tokenizer_config.json: 100% 42.0/42.0 [00:00<00:00, 1.12kB/s]
```

```
source.spm: 100% 778k/778k [00:00<00:00, 11.9MB/s]
```

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target.spm: 100% 802k/802k [00:00<00:00, 36.6MB/s]
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downl

```
vocab.json: 1.34M/? [00:00<00:00, 33.9MB/s]
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/arian/tokenization_arian.py:176: UserWarning: Recommended: pip install sacremoses.
  warnings.warn("Recommended: pip install sacremoses.")
```

```
pytorch_model.bin: 100% 301M/301M [00:02<00:00, 178MB/s]
```

```
Loading weights: 100% 258/258 [00:00<00:00, 401.94it/s, Materializing param=model.shared.weight]
```

```
model.safetensors: 100% 301M/301M [00:03<00:00, 149MB/s]
```

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but both are present in the checkpoints, s
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but both are present in the checkpoints, s

```
generation_config.json: 100% 293/293 [00:00<00:00, 11.7kB/s]
```

```
Translator for Helsinki-NLP/opus-mt-en-fr initialized successfully.
```

```
print(translator_fr("I love artificial intelligence"))
```

```
J'aime l'intelligence artificielle
```

```
# Manual setup for English to German translation (due to pipeline KeyError)
model_name_de = "Helsinki-NLP/opus-mt-en-de"
tokenizer_de = AutoTokenizer.from_pretrained(model_name_de)
model_de = AutoModelForSeq2SeqLM.from_pretrained(model_name_de)

def translate_en_to_de(text):
    inputs = tokenizer_de(text, return_tensors="pt", padding=True, truncation=True)
    outputs = model_de.generate(**inputs)
    return tokenizer_de.decode(outputs[0], skip_special_tokens=True)

translator_de = translate_en_to_de
print(f"Translator for {model_name_de} initialized successfully.")

# Manual setup for English to Hindi translation (due to pipeline KeyError)
model_name_hi = "Helsinki-NLP/opus-mt-en-hi"
tokenizer_hi = AutoTokenizer.from_pretrained(model_name_hi)
model_hi = AutoModelForSeq2SeqLM.from_pretrained(model_name_hi)

def translate_en_to_hi(text):
    inputs = tokenizer_hi(text, return_tensors="pt", padding=True, truncation=True)
    outputs = model_hi.generate(**inputs)
    return tokenizer_hi.decode(outputs[0], skip_special_tokens=True)

translator_hi = translate_en_to_hi
print(f"Translator for {model_name_hi} initialized successfully.")
```

```

config.json:      1.33k/? [00:00<00:00, 19.2kB/s]

tokenizer_config.json: 100%                                42.0/42.0 [00:00<00:00, 938B/s]

source.spm: 100%                                           768k/768k [00:00<00:00, 7.34MB/s]

target.spm: 100%                                           797k/797k [00:00<00:00, 11.7MB/s]

vocab.json:      1.27M/? [00:00<00:00, 7.41MB/s]
/usr/local/lib/python3.12/dist-packages/transformers/models/arian/tokenization_arian.py:176: UserWarning: Recommended: pip install sacremoses.
  warnings.warn("Recommended: pip install sacremoses.")

pytorch_model.bin: 100%                                    298M/298M [00:03<00:00, 130MB/s]

model.safetensors: 100%                                    298M/298M [00:05<00:00, 76.6MB/s]

Loading weights: 100%                                     258/258 [00:00<00:00, 549.17it/s, Materializing param=model.shared.weight]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but both are present in the checkpoints, s
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but both are present in the checkpoints, s
generation_config.json: 100%                               293/293 [00:00<00:00, 23.1kB/s]

Translator for Helsinki-NLP/opus-mt-en-de initialized successfully.

config.json:      1.39k/? [00:00<00:00, 60.7kB/s]

tokenizer_config.json: 100%                                44.0/44.0 [00:00<00:00, 1.70kB/s]

source.spm: 100%                                           812k/812k [00:00<00:00, 3.52MB/s]

target.spm: 100%                                           1.07M/1.07M [00:00<00:00, 2.82MB/s]

vocab.json:      2.10M/? [00:00<00:00, 7.66MB/s]

pytorch_model.bin: 100%                                    306M/306M [00:08<00:00, 34.3MB/s]

Loading weights: 100%                                     258/258 [00:00<00:00, 432.01it/s, Materializing param=model.shared.weight]

model.safetensors: 100%                                    306M/306M [00:03<00:00, 146MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but both are present in the checkpoints, s
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but both are present in the checkpoints, s
generation_config.json: 100%                               293/293 [00:00<00:00, 12.2kB/s]

Translator for Helsinki-NLP/opus-mt-en-hi initialized successfully.

```

```

sentences = [
    "I love learning new technologies.",
    "The weather is very pleasant today.",
    "Artificial Intelligence is transforming industries.",
    "Can you help me with this problem?",
    "She went to the market and bought fruits.",
    "Although it was raining, we continued the match.",
    "Data science requires statistics and programming knowledge.",
    "What are you doing right now?",
    "He completed the task before the deadline.",
    "Social media influences public opinion.",
    "Machine learning models require proper training.",
    "The results were surprising and unexpected.",
    "Do you understand this concept clearly?",
    "The system failed due to a technical error.",

```

```
"We should work together to solve this issue."
]
```

```
results = []

for sentence in sentences:
    fr = translator_fr(sentence)
    de = translator_de(sentence)
    hi = translator_hi(sentence)

    results.append((sentence, fr, de, hi))
```

```
for res in results:
    print("\nOriginal:", res[0])
    print("French:", res[1])
    print("German:", res[2])
    print("Hindi:", res[3])
```

Original: I love learning new technologies.
 French: J'adore apprendre les nouvelles technologies.
 German: Ich liebe es, neue Technologien zu lernen.
 Hindi: मैं नई तकनीक सीखने को प्यार करता हूँ.

Original: The weather is very pleasant today.
 French: Le temps est très agréable aujourd'hui.
 German: Das Wetter ist heute sehr angenehm.
 Hindi: मौसम आज बहुत सुखद है.

Original: Artificial Intelligence is transforming industries.
 French: L'intelligence artificielle transforme les industries.
 German: Künstliche Intelligenz verändert die Industrie.
 Hindi: कलात्मक तत्व उद्योगों को परिवर्तित कर रहा है.

Original: Can you help me with this problem?
 French: Pouvez-vous m'aider avec ce problème ?
 German: Können Sie mir bei diesem Problem helfen?
 Hindi: आप इस समस्या के साथ मेरी मदद कर सकते हैं?

Original: She went to the market and bought fruits.
 French: Elle est allée au marché et a acheté des fruits.
 German: Sie ging auf den Markt und kaufte Früchte.
 Hindi: वह बाजार में गया और फल लाया.

Original: Although it was raining, we continued the match.
 French: Même s'il pleuvait, nous avons continué le match.
 German: Obwohl es regnete, setzten wir das Spiel fort.
 Hindi: हालाँकि बारिश हो रही थी, फिर भी हमने मैच जारी रखा ।

Original: Data science requires statistics and programming knowledge.
 French: La science des données exige des statistiques et des connaissances en matière de programmation.
 German: Die Datenwissenschaft erfordert Statistiken und Programmierkenntnisse.
 Hindi: डाटा विज्ञान के लिए सांख्यिकी और प्रोग्रामिंग ज्ञान की जरूरत है.

Original: What are you doing right now?
 French: Qu'est-ce que tu fais en ce moment ?
 German: Was machst du gerade?
 Hindi: तुम अभी क्या कर रहे हो?

Original: He completed the task before the deadline.
French: Il a terminé la tâche avant la date limite.
German: Er hat die Aufgabe vor Ablauf der Frist erledigt.
Hindi: उसने इस कार्य को मृत रेखा के पहले पूरा किया ।

Original: Social media influences public opinion.
French: Les médias sociaux influencent l'opinion publique.
German: Soziale Medien beeinflussen die öffentliche Meinung.
Hindi: सामाजिक मीडिया जनता की राय पर असर करता है ।

Original: Machine learning models require proper training.
French: Les modèles d'apprentissage automatique nécessitent une formation appropriée.
German: Maschinelle Lernmodelle erfordern ein angemessenes Training.
Hindi: मशीन सीखने के मॉडल सीखने के लिए उचित प्रशिक्षण की ज़रूरत है।

Original: The results were surprising and unexpected.

```
import pandas as pd
```

```
df = pd.DataFrame(results, columns=["English", "French", "German", "Hindi"])
```